

Tucatinib, trastuzumab and capecitabine

Tucatinib (Tukysa[®]), trastuzumab and capecitabine is a cancer drug treatment. It is used to treat breast cancer.

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What is tucatinib, trastuzumab and capecitabine?

Tucatinib (Tukysa[®]), trastuzumab and capecitabine is a combination of cancer drugs. It used to treat [breast cancer](#). It may sometimes be used to treat other cancers.

Tucatinib is also called Tukysa[®]. Tucatinib and trastuzumab are targeted therapy drugs. Capecitabine is a type of chemotherapy.

It is best to read this information with our general information about [targeted therapy drugs](#), [chemotherapy](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

DPD testing before treatment

Before starting this treatment, you should have a blood test to check for gene changes that affect how your body produces DPD. DPD is an enzyme that helps the body break down capecitabine.

If you have low DPD levels or no DPD (DPD deficiency), you have more risk of serious or life-threatening side effects from capecitabine. To reduce your risk, your doctor may give you:

- a lower dose of capecitabine to start with
- a different type of chemotherapy.

Having low DPD levels does not usually cause any symptoms. Without a test, you will not know whether your DPD levels are low. Testing finds most people who are affected, but not all. Your doctor, nurse or pharmacist can give you more information about your risk of DPD deficiency, before you start treatment.

How tucatinib, trastuzumab and capecitabine is given

During a course of treatment, you will see someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

You will have blood tests before and regularly during your treatment. This is to check:

- that your blood cells are at a safe level for you to have chemotherapy
- how well your liver and kidneys are working.

A doctor, nurse or pharmacist will see you before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your treatment.

Your course of treatment

You usually have this treatment as a course of several cycles of treatment. Each cycle of treatment is 21 days. You have the treatment in the following ways:

- **On day 1** - you have trastuzumab as a drip into a vein (intravenous infusion) or as an injection under the skin (subcutaneous). You also start taking tucatinib and capecitabine tablets.
- **On days 1 to 14** – you take tucatinib and capecitabine tablets.
- **On days 15 to 21** - you take just tucatinib tablets.

Your cancer team will discuss your treatment plan with you. They may give you a copy of the treatment plan to take home.

They will give you the tucatinib and capecitabine tablets to take home. Always take them exactly as explained. This is important to make sure they work as well as possible for you.

You may be given tablets of different strengths to make up your dose. Depending on the side effects, your dose of tucatinib or capecitabine may change from one cycle to the next. It is very important that you know how many tablets to take each time you start a new cycle.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your medicines exactly as they have been explained to you.

Having trastuzumab

On day 1 of each cycle, you have trastuzumab. Usually, trastuzumab is given in a chemotherapy day unit or clinic as an outpatient. It can be given as a drip into a vein (intravenous infusion) or as an injection under the skin (subcutaneous). Your cancer team will explain how you will have it. You will have it in 1 of the following ways:

- **As a drip** - the nurse gives you the first dose (infusion) slowly. This is usually over 1½ hours (90 minutes). For the first infusion, you may stay in hospital for a few hours so they can monitor you for any reaction. If the first infusion does not cause a reaction, you can have the following infusions over 30 to 90 minutes. You go home about 1 hour after the treatment ends.
- **As an injection under the skin** - the nurse gives you the injection under the skin on your thigh. This takes a few minutes. After the first injection, the nurse monitors you for 30 minutes to make sure you do not have a reaction. If the first injection does not cause a reaction, you can go home sooner after the following injections.

Taking tucatinib and capecitabine tablets

You take tucatinib and capecitabine 2 times a day. On days 1 to 14, you can take both drugs at the same time if you want to.

You can take tucatinib with or without food. But you should take capecitabine within 30 minutes after eating a meal. Try to take your tablets at the same times each day. For example, you could take them in the morning after breakfast and then 10 to 12 hours later, after your evening meal or a snack.

Do not chew, crush, break or cut your tablets. Swallow them whole with a glass of water. Tell your doctor, nurse or pharmacist if you find it difficult to swallow the tablets.

If you forget to take the tablets, you should take your next dose at your regular time. Do not take a double dose to make up for the missed dose.

Other things to remember about your tablets:

- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, somewhere children cannot see or reach them.
- If you are sick just after taking the tablets, do not take an extra dose. Just take your next dose at the usual time.
- If your treatment is stopped, return any unused tablets to the pharmacist.
- Wash your hands after taking your tablets.
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Other people should avoid direct contact with them.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

If your body does not produce enough of an enzyme called DPD, you may have a higher risk of severe or life-threatening side effects.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while trastuzumab is being given

Allergic reaction

Some people have an allergic reaction to trastuzumab while they are having it. The first dose is the most likely to cause a reaction. If you are having trastuzumab as an infusion, it is usually given more slowly than later treatments.

Before treatment, you may be given medicines to help prevent or reduce any reaction. Signs of a reaction can include:

- feeling hot or flushed
- shivering

- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy (abdomen) or chest.

Your nurse will check you for signs of a reaction during and after your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any signs or feel unwell after you get home, contact the hospital straight away.

Injection site

If you have trastuzumab as an injection under the skin, the skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin.

If this happens, let your doctor know. Painkillers may help. The nurse will swap between your left and right thigh.

Common side effects

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Diarrhoea

This treatment may cause severe [diarrhoea](#). Diarrhoea means passing more stools (poo) than is usual for you, or having watery or loose stools. If you have a stoma, it will be more active than usual.

Your hospital team may give you anti-diarrhoea drugs to take at home.

If you have diarrhoea or an increase in stoma activity:

- follow any advice from your hospital team about taking anti-diarrhoea drugs
- drink at least 2 litres (3½ pints) of fluids each day
- avoid alcohol, caffeine, milk products, high-fat foods and high-fibre foods.

Contact the hospital straight away if:

- you have tummy pain or symptoms of infection (see 'Risk of infection' above)
- you have diarrhoea at night
- you have diarrhoea more than 4 times in a day
- you have a moderate or severe increase in stoma activity
- the anti-diarrhoea drugs do not work within 24 hours.

Your doctor may ask you to stop taking the tucatinib or capecitabine tablets. When the diarrhoea is better, they will tell you if you can start taking it again. Sometimes they reduce the dose.

Effects on the liver

This treatment may affect how your liver works. You will have blood tests before your treatment and every 3 weeks during treatment to check how well your liver is working.

Sometimes liver changes can be serious. Contact the hospital straight away if you have any of these symptoms during treatment or after it finishes:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- itching
- pain in the upper right side of your tummy
- loss of appetite or feeling sick.

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Effects on the heart

This treatment can affect how the heart works. You will have tests to check how well your heart is working. These will be done before you start treatment, and at regular times during and after treatment.

If the treatment is causing heart problems, your doctor may make changes to your treatment.

Contact the hospital straight away on the 24-hour number they have given you if you have any of these symptoms during or after treatment:

- breathlessness or a cough
- swollen feet and ankles
- dizziness
- changes to your heartbeat.

Stop taking the capecitabine. Do not start taking it again until your doctor, nurse or pharmacist tells you to.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

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- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Hand-foot (palmar-plantar) syndrome

This treatment can affect the palms of your hands and the soles of your feet. This is called palmar-plantar or hand-foot syndrome.

If you have white skin these areas may become red. If you have black or brown skin, these areas might get darker.

The skin on the palms of your hands and the soles of your feet may:

- be sore
- be painful, tingle, or swell
- peel, crack or blister.

If you have any of these symptoms, contact the hospital straight away on the 24-hour number. They can give you advice. This is especially important if you have any broken skin or if walking is difficult. They can prescribe creams and painkillers to help.

You can care for your hands and feet by:

- keeping your hands and feet cool by washing in cool water
- gently moisturising your hands and feet regularly
- wearing gloves to protect your hands and nails when working in the house or garden
- wearing loose cotton socks and avoiding tight-fitting shoes and gloves.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco

- avoid food or drinks that irritate your mouth and throat.

Skin and nail changes

This treatment may affect your [skin and nails](#). It may cause a rash, which might be itchy. Your doctor, nurse or pharmacist can tell you what to expect. If your skin feels dry, try using an unperfumed moisturising cream every day.

Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice and may give you creams or medicines to help. Any changes to your skin are usually temporary and improve when treatment finishes.

Very rarely, capecitabine may cause a more serious skin condition. You may have a skin rash which then blisters and your skin can peel. You may also feel unwell with flu-like symptoms such as a high temperature and joint pain. If you have any of these symptoms, stop taking the capecitabine tablets until you have spoken to your doctor. Call your doctor or the 24-hour number the hospital has given you. If you cannot get through, call the **NHS urgent advice** number which is **111**.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Less common side effects

Joint and back pain

You may get pain in your joints or other areas, such as your back, during treatment. If this happens, tell your doctor so they can give you painkillers. Tell them if the pain does not get better. Having warm baths and taking regular rests may help.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Hair loss

Your hair may get thinner. But you are unlikely to lose all the hair from your head. Hair loss usually starts after your first or second treatment. It is almost always temporary, and your hair will usually grow back after treatment ends.

Your nurse can talk to you about ways to cope with [hair loss](#).

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Dizziness

Sometimes capecitabine may cause dizziness. Let your doctor know if you feel dizzy. It is important not to drive or operate machinery if you notice these effects.

Mood and sleep changes

You may have some [mood changes](#) during this treatment. You may feel low, depressed or anxious. You may have [trouble sleeping](#). Tell your doctor or nurse if you notice any changes.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Sore eyes

Your eyes may become watery and feel sore. Your doctor can prescribe eye drops to help with this. If your eyes become red and inflamed, tell your doctor. This is called conjunctivitis. You may need eye drops to treat it.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they

protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex during this treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert health professionals and people living with cancer.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 January 2024 | Next review: 01 January 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

Get in touch



0300 1000 200

Our Support Line is available Monday to Friday, 9:30am-4:30pm



Email us

Email the Macmillan Fundraising team if you have any questions.



Fundraising FAQs

Answers to common questions our supporters have about supporting Macmillan – how to pay in money, changing your details, making a donation.



Fundraising tips

Get top tips and advice from Macmillan on how to maximise your fundraising.